

AKSHAY JAMBEKAR

Python Developer | ML / Data Engineer

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PROFESSIONAL SUMMARY

Python Developer and Data Engineer with 4+ years of IT experience spanning ML model development, Python ETL pipelines, Oracle database administration, and Snowflake-based analytics. Delivered production ML solutions in logistics (Reinforcement Learning, Gradient Boosting) and NLP. Currently building expertise in cloud-native data engineering — Apache Airflow orchestration, Docker, AWS, and MLOps infrastructure. Actively transitioning into AI infrastructure and Data Engineering roles.

PORTFOLIO PROJECT

DE Pipeline — ETL Pipeline with Airflow Orchestration | github.com/akshayjambekar/DE_Pipeline

- Built end-to-end ETL pipeline using Python, Apache Airflow, and PostgreSQL (Neon cloud)
- Implemented Bronze layer with date-partitioned Parquet storage and SCD Type 2 change tracking
- Orchestrated 4-task Airflow DAG with retries, idempotent execution via `{{ ds }}`, and structured logging
- Containerized Airflow environment using Docker Compose; deployed on GitHub Codespaces
- **Tools:** Python, Apache Airflow, Docker, PostgreSQL, Parquet, GitHub

MACHINE LEARNING PROJECTS

Trailer Space Optimization — Reinforcement Learning

- Designed and benchmarked a Reinforcement Learning model (PPO Maskable) to evaluate trailer space utilization and SKU split decisions against existing heuristics
- Diagnosed that heuristic performance matched RL output due to upstream data ordering in a stored procedure — identifying the root cause beyond the model
- Evaluated alternatives including OR-Tools and First Fit Decreasing bin packing as part of a structured solution comparison
- **Stack:** Python, Stable-Baselines3, OR-Tools, Pandas

Customer Sentiment Analysis — NLP Pipeline

- Designed and implemented NLP pipeline to analyze customer reviews and extract sentiment scores
- Applied lemmatization, BOW, TF-IDF, and GloVe embeddings; generated actionable insights for product and marketing teams
- **Stack:** Python, NLTK, Scikit-learn, TF-IDF, GloVe, Pandas

WORK EXPERIENCE

Accenture

Nov 2024 – Present

Senior Software Analyst | Python Developer

- Designed AI/ML models in Python to optimize logistics operations including trailer space utilization and pallet forecasting
- Built end-to-end ETL pipelines ingesting Excel data into Snowflake, eliminating ~8 hours/week of manual effort
- Designed Python-based Streamlit UI for operational reporting, reducing report generation time by ~50%
- Migrated 10+ legacy MS Access/VBA workflows to Python and SQL; enhanced Angular front-end components for improved UX
- Performed data manipulation using Pandas and NumPy across 500K+ row datasets for analytics use cases

Capgemini

Jun 2021 – Oct 2024

Senior Analyst | Python Developer

- Developed e-commerce delivery prediction model (binary classification) to improve logistics planning performance
- Built NLP sentiment analysis pipeline on customer reviews to generate product improvement insights
- Administered Oracle Exadata platforms supporting enterprise analytics across 15+ production databases
- Resolved high-priority production incidents within SLA; maintained 99.9% uptime through patching and failover operations

TECHNICAL SKILLS

Languages: Python, SQL, HTML, CSS

ML / AI: Scikit-learn, XGBoost, Stable-Baselines3, NLTK, ANN, CNN, NLP (TF-IDF, GloVe), MLOps (in progress)

Data Engineering: Apache Airflow, Docker, Parquet, Pandas, NumPy, Snowflake, PostgreSQL, Oracle Exadata

Backend / Frameworks: Django, FastAPI, DjangoRestFramework, Streamlit, Angular

Cloud & Infra: AWS (learning), Azure ML, Git, GitHub, Linux

EDUCATION

Bachelor of Engineering – Computer Engineering

2016 – 2020

Savitribai Phule Pune University | Distinction – 77.12%

CERTIFICATIONS

- Applied Machine Learning in Python – Coursera
- Data Science & Artificial Intelligence – Boston Institute of Analytics
- Machine Learning Practical Workout | 8 Real-World Projects – Udemy
- Python & SQL Certified – HackerRank